Week 1 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse.

W01-Q01-PA

1. Write A and B explicitly before applying any set operation.
2. The complement contains the elements of Ω that are not in A. To disprove $B \subseteq A$, one element of $B \quad A$ is enough.

W01-Q01-PB

1. Compare $f(-1)$ with $f(1)$, then check whether every element of the stated codomain occurs as an output.
2. For the reverse direction of the parity claim, suppose n^2 is even and write a contrary odd integer as $n = 2k+1$.

W01-Q02-PA

1. Negating a universal statement changes $\forall$ to $\exists$.
2. The negation of $x^2 + 1 > 0$ is $x^2 + 1 \leq 0$. Use the sign of x^2 for real x to assess the original statement.

W01-Q02-PB

1. Verify the formula at $n = 1$, then assume it holds at an arbitrary $n \geq 1$.
2. The sum at $n+1$ is the sum at n plus $3(n+1)-1$. Substitute the induction hypothesis and factor the result into the required form.

W01-Q03-PA

1. Both logarithm arguments must be positive.
2. On the admissible domain, use the product rule for logarithms and the injectivity of log to obtain a quadratic equation. Check both roots against the original domain.

W01-Q03-PB

1. Expand the product over $i = 1, 2, 3, 4$; powers with the same base multiply by adding exponents.
2. The benefits form a geometric series with first term 100 and common ratio $3/4$. For $t = 0, \ldots, N$, use the finite geometric-sum formula.
3. Reduce the threshold condition to $(3/4)^{N+1} \leq 1/8$. When dividing by $\log(3/4) < 0$, remember what happens to the direction of the inequality.

W01-Q04-PA

1. The argument of a real logarithm must be strictly positive.
2. Because log is strictly increasing, compare $9-x^2$ directly with 5, then intersect the solution with the domain.

W01-Q05-PA

1. Apply the product rule to $x^2 \log x$; the domain $x > 0$ permits the derivative of $\log x$.
2. For the integral, integration by parts gives an antiderivative of $\log x$. Evaluate it at both endpoints.

W01-Q05-PB

1. A derivative is a local rate of change. Calculate $C'(q)$ and attach units to $C'(6)$.
2. A change of one whole unit is $C(7) - C(6)$, not automatically $C'(6)$. Calculate both before assessing the word "exactly".

W01-Q06-PA

1. At the centre $a = 1$, record $f(1)$, $f'(1)$ and $f''(1)$.
2. Use $T_2(x) = f(1) + f'(1)(x-1) + \frac{1}{2}f''(1)(x-1)^2$, and substitute $x - 1 = 0.2$.

W01-Q06-PB

1. Compute each absolute error relative to the supplied comparison value.
2. One comparison near one centre can show which approximation is better at that point; consider why it cannot establish a global rule by itself.

W01-Q07-PA

1. Find the stationary point from $\pi'(q) = 0$ and check that it is feasible.
2. A global check on a closed interval compares the objective at every interior candidate and both endpoints. The sign of $\pi''(q)$ supplies additional curvature information.

W01-Q07-PB

1. A stationary point alone need not beat endpoints or even be a maximum.
2. Use both the endpoint comparison from Part A and the strict concavity of this quadratic to explain why the conclusion is valid here.

W01-Q08-PA

1. Start with inclusion–exclusion: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
2. "Neither" is the complement of the union. "Exactly one" consists of the union after removing the intersection once.

W01-Q08-PB

1. The sum $\mathbb{P}(A) + \mathbb{P}(B)$ counts outcomes in $A \cap B$ twice.
2. Subtract the overlap once so that every outcome in the union is counted exactly once.

W01-Q09-PA

1. Without the restriction, choose an unordered three-person panel and then choose its chair.
2. Count the forbidden outcomes by fixing Ada and Bruno on the panel, choosing the third member, and then choosing one of the three members as chair. Subtract these from the unrestricted count.

W01-Q10-PA

1. A four-proposal selection representing three fields has composition $2, 1, 1$. Count separately according to which field supplies two.
2. For the complement, define events for "no theory", "no applied economics" and "no economic history". Apply inclusion–exclusion to their union, including the possible pairwise intersections.